

The Marauder's Ledger

AI-Powered Financial Anomaly Detection

"I solemnly swear that I am up to no good."

React 19

FastAPI

ML Ensemble

Gemini AI

VectorAI

Docker

Complete Workflow Documentation

1. Project Overview

The Marauder's Ledger (also branded as **OmniLedger**) is a full-stack AI-powered web application that detects financial anomalies in bank transaction CSV files. Users upload their transaction data, and the system runs a sophisticated 10-model machine learning ensemble to identify suspicious patterns, visualizes them on an interactive Marauder's Map-themed SVG interface, generates AI-powered narratives via Google Gemini, and reads them aloud via ElevenLabs text-to-speech.

Technology Stack

Frontend

- React 19 + TypeScript 6.0
- Vite 8 (build tool)
- React Router v7 (routing)
- TanStack Query v5 (server state)
- Axios (HTTP client)
- Tailwind CSS v4 (styling)
- Framer Motion (animations)
- Recharts (charts)
- react-dropzone (CSV upload)

Backend & Infrastructure

- Python 3.11 + FastAPI + Uvicorn
- Actian VectorAI + SQLite (databases)
- Celery + Redis (background tasks)
- scikit-learn, XGBoost, LightGBM, CatBoost
- Google Gemini 2.0 Flash (AI narrative)
- ElevenLabs API (text-to-speech)
- JWT + bcrypt (authentication)
- Docker Compose (5 services)

Project Structure

The project contains two frontend applications:

- **frontend/** — Harry Potter themed (9 pages: Landing, Dashboard, AnomalyDetail, MischiefList, Vault, Pensieve, OwlPost, Profile, Admin)
- **stitch-frontend/** — OmniLedger tech/industrial themed (18+ pages including Fleet Management, Access Control, Global Feed, Integrations, Performance)

Both frontends share the same **backend/** FastAPI application, ML engine, and database layer.

2. Authentication & User Onboarding Workflow



Authentication Flow Details

Registration

- POST /api/auth/register with name, email, password
- Password hashed with bcrypt (hash_password)
- User stored in SQLite users table
- Returns JWT + user info on success

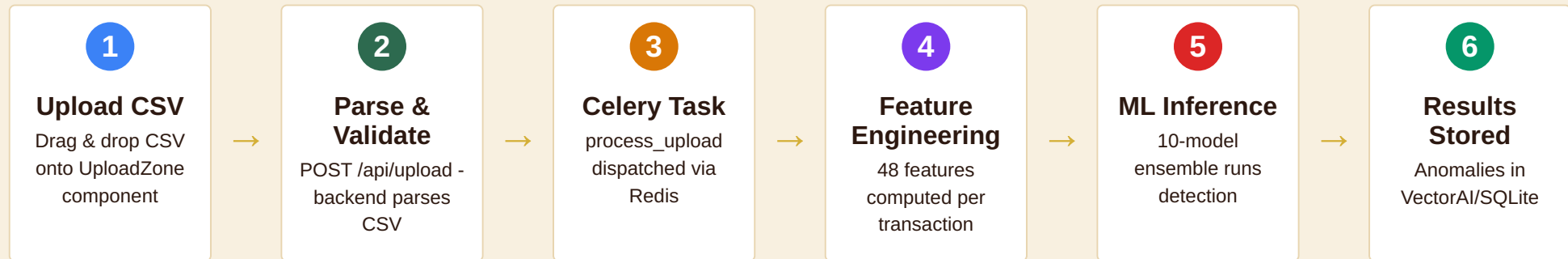
Login

- POST /api/auth/login with email, password
- Verifies against bcrypt hash
- Returns JWT with sub (user_id), email, name
- Token expires in 24 hours

Session Management

- JWT stored in localStorage key 'marauders_token'
- User object stored in localStorage key 'marauders_session'
- Axios interceptor adds 'Authorization: Bearer ' to all requests
- ProtectedRoute component redirects to /login if not authenticated
- 401 response from any API auto-clears session and redirects
- Rate limiting: 30 requests per 60s per IP on auth endpoints
- Backend verifies via FastAPI get_current_user dependency (HTTPBearer)
- User ownership checked on all endpoints (403 if mismatch)

3. CSV Upload & Processing Workflow



Upload Details

CSV Requirements

- Required columns: amount, category, merchant, hour, day
- Uploaded via react-dropzone on the Landing page
- File sent to backend via FormData
- Backend parses with pandas (pd.read_csv)

Processing Pipeline

- Celery task: process_upload → parse → features → detect
- Frontend polls GET /api/batches/:id/progress every 1s
- Polling timeout: 60 seconds
- On completion → navigate to /dashboard with toast
- Synchronous fallback if Celery/Redis unavailable

4. ML Anomaly Detection Engine (Deep Dive)

Model Architecture

Model	Type	Weight / Role
Random Forest	Supervised Ensemble	0.15 — Core ensemble member
Gradient Boosting	Supervised Ensemble	0.15 — Core ensemble member
XGBoost	Gradient Boosting	0.25 — High-performance learner
LightGBM	Gradient Boosting	0.25 — High-performance learner
CatBoost	Gradient Boosting	0.10 — Supplementary
Extra Trees	Supervised Ensemble	0.10 — Supplementary
Isolation Forest	Unsupervised	Feature — Outlier score
Local Outlier Factor	Unsupervised	Feature — Density score
One-Class SVM	Unsupervised	Feature — Boundary score
Rule-Based Scoring	Heuristic	0.25 weight — Domain rules

48 Engineered Features

Base Features (35)

- Amount: raw, log-transformed, Z-score
- Category: one-hot encoded
- Merchant: encoded, new merchant flag
- Hour: raw, cyclical sin/cos encoding
- Day: raw, cyclical sin/cos, weekend flag
- Rolling: mean, std, count (7-day windows)
- Interaction: category×amount, hour×day
- Velocity: transaction count in time window

ML & Rule Scores (13)

- Isolation Forest anomaly score
- Local Outlier Factor score
- One-Class SVM decision function
- Rule Score: heuristic anomaly indicator
- Amount threshold violation score
- Category spending spike score
- Unusual hour indicator
- New merchant detection score
- Rolling average deviation score

Scoring Formula

final_score = 0.75 x ensemble_score + 0.25 x rule_score

- ensemble_score = weighted average of 9 ML model predictions
- rule_score = heuristic-based anomaly indicators (0-1)
- Threshold: final_score > 0.5 → flagged as anomaly

• Severity: **Peeves (low)** **Boggart (medium)** **Dementor (high)**

Feature scaling via StandardScaler. All models serialized with joblib.

5. Dashboard & Marauder's Map Workflow

After CSV processing completes, the user is redirected to the Dashboard where results are visualized on the interactive Marauder's Map.

Map Locations (Spending Categories)

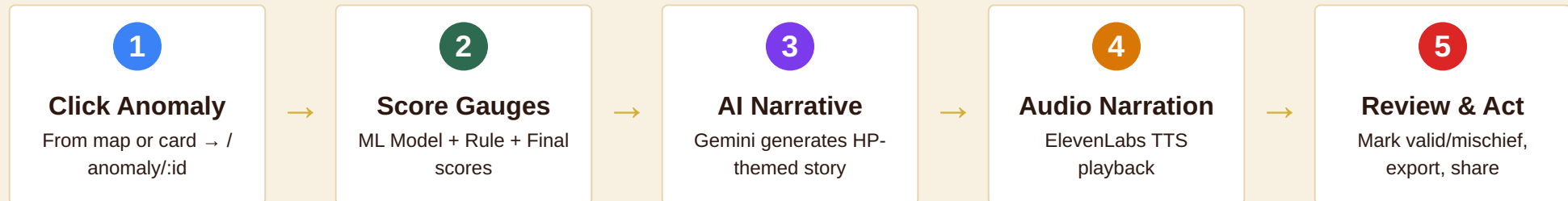
Location	Category	Color
Hogwarts	Food & Dining	Purple
Hogsmeade	Shopping	Blue
Gringotts	Bills & Utilities	Gold
Diagon Alley	Entertainment	Teal/Green
Platform 9 3/4	Travel	Red

Interactive Features

- **Animated Footprints** — Transactions appear as walking footprints traversing the map between locations
- **Anomaly Pulsing** — Detected anomalies glow red and pulse to draw immediate attention
- **Severity Colors** — Peeves (green/low), Boggart (amber/medium), Dementor (red/high)
- **Click to Inspect** — Click any footprint or anomaly to navigate to the detail page
- **Category Filter Tabs** — Moony (Food), Wormtail (Shopping), Padfoot (Bills), Prongs (All)
- **Spending Trend Chart** — Bottom panel shows Recharts line chart of spending over time
- **Stats Bar** — Top bar shows total anomalies, high/medium/low counts, total financial impact
- **Right Sidebar** — Anomaly cards list with severity badges

The MaraudersMap component (257 lines) is an SVG-based interactive visualization built with React and Framer Motion for animations. The SpendChart component uses Recharts for the spending trend line chart.

6. Anomaly Investigation Workflow



Anomaly Detail Page Components

Investigation Tools

- **ScoreGauge (x3)** — Circular gauges showing ML Model Score, Rule Score, and Final Mischief Score (0-100%)
- **SeverityBadge** — Color-coded severity indicator (Dementor/Boggart/Peeves)
- **NarrativeCard** — Gemini-generated AI narrative with typewriter animation effect
- **AudioPlayer** — Waveform audio player for TTS narration (play/pause controls)

Actions & Context

- **Status Management** — Toggle between 'Valid' (confirmed legitimate) and 'Mischief' (confirmed fraud)
- **Triggered Rules** — Tags showing which rules the anomaly violated
- **Related Transactions** — Nearby transactions for context and comparison
- **Export & Share** — Export HTML report via pdfExport.ts, share link via clipboard

Narrative Generation Flow

1. Frontend calls GET /api/narratives/:id
2. Backend checks cache (existing narrative in DB)
3. If not cached: calls Google Gemini 2.0 Flash API with anomaly data
4. Gemini generates Harry Potter-themed narrative explaining the anomaly
5. Narrative stored in database for future requests
6. Frontend displays with typewriter animation
7. Graceful fallback: static template if Gemini API unavailable

TTS Audio Flow

1. User clicks play on AudioPlayer widget
2. Frontend calls GET /api/narratives/:id/audio
3. Backend checks cache for existing MP3 audio
4. If not cached: calls ElevenLabs TTS API (voice: Roger)
5. Audio streamed back as MP3 (Content-Type: audio/mpeg)
6. Sequential request locking per anomaly ID prevents duplicates
7. Graceful fallback: narrative displayed as text only

7. Voice Chat & Power User Features

Voice Chat Assistant

- Persistent FAB (floating action button) in bottom-right corner of every page
- Supports voice input via Web Speech API (SpeechRecognition)
- Text input fallback for environments without microphone access
- Sends POST /api/chat/message with page context (anomaly_id, batch_id)
- Gemini responds in Marauder's Map persona character
- ElevenLabs streams audio response back (X-Chat-Response header + audio/mpeg)
- Chat history preserved during session
- Example queries: "What anomalies were detected?", "Explain this transaction", "Summarize my spending"

Command Palette (Cmd+K / Ctrl+K)

- Quick navigation to any page in the application via fuzzy search
- Search transactions, anomalies, and investigations
- Toggle dark mode directly from command palette
- Access keyboard shortcuts reference modal
- Context-aware commands based on the current page

Dark Mode & Theming

- Full dark/light theme toggle via DarkModeToggle component
- Persistent theme preference stored in localStorage ('theme' key)
- Seamless transitions via Tailwind CSS dark mode classes
- Parchment aging effect in the Harry Potter theme
- Consistent design across all 18+ pages

PDF Export & Sharing

- Export anomaly reports as formatted HTML documents

- Share anomaly details via clipboard link
- PDF includes: score gauges, AI narrative, triggered rules, transaction details
- Export function in AnomalyDetail page via pdfExport.ts service

8. Supporting Pages Workflow

Page	Route	Purpose	Key Features
MischiefList	/ledger	Historical anomaly table	Search, filter by severity, sort by date, paginated list
Vault	/vault	Gringotts-style spending overview	Total balance, category breakdown with ProgressRing, recent anomalies
Pensieve	/pensieve	Deep spending analysis	Spending trend line charts, time range selectors, risk breakdown pie chart
Owl Post	/owl-post	Notification center	Read/unread anomaly alerts, categorized by severity with timestamps
Profile	/profile	Wizard's dossier	User stats, spending by category rings, recent cases list
Admin	/admin	System settings	Theme toggle, parchment aging, mischief taxonomy editor, API status dashboard

Navigation

- Persistent SideBarNav component on all pages
- Header with UserMenu dropdown (profile, settings, logout)
- Command Palette (Cmd+K) for quick page navigation
- Keyboard shortcut help modal
- PageTransition animations via Framer Motion AnimatePresence

State Management

- **React Context:** AuthContext (user/JWT), AppContext (batchId, userId), ToastContext (notifications)
- **TanStack Query:** All API data managed via useQuery + useMutation with automatic caching and invalidation
- **Query Keys:** ['anomalies'], ['narrative', id], ['audio', id], ['spending-day'], ['spending-category'], ['batches'], ['transactions']
- **localStorage:** marauders_token, marauders_session, marauders_onboarded, theme

9. End-to-End Data Flow Architecture

Layer	Technology	Responsibilities
User Browser	React 19 + TypeScript	BrowserRouter → AuthProvider → QueryClientProvider → AppProvider. TanStack Query for all API data. Components: SidebarNav, UploadZone, MaraudersMap, VoiceChatWidget
API Gateway	FastAPI + Uvicorn	17+ REST endpoints. JWT verification via <code>get_current_user</code> . CORS middleware, rate limiting (30 req/60s per IP). Async handlers with HTTPX for external API calls.
Background Processing	Celery + Redis	<code>process_upload</code> : parse CSV → feature engineering → ML inference. Frontend polls progress every 1s. Synchronous fallback if unavailable.
ML Engine	scikit-learn + XGBoost + LightGBM + CatBoost	10 models: 6 supervised + 3 unsupervised + 1 rule engine. 48 features per transaction → scaled → ensemble prediction. Hybrid score: $0.75 * \text{ML} + 0.25 * \text{rules}$.
Data Layer	Action VectorAI + SQLite	VectorAI: primary vector database with hybrid search. SQLite: automatic fallback + user auth storage. Auto-detection of server capabilities.
External AI	Gemini 2.0 Flash + ElevenLabs	Gemini: narrative generation + voice chat responses. ElevenLabs: TTS narration streaming MP3. Thread-safe locking per anomaly. Graceful degradation.

Deployment Architecture (Docker Compose)

Service	Role	Port
Frontend	React 19 + Nginx static serving	3000
Backend	FastAPI + Uvicorn ASGI server	8000
Redis	Celery message broker + cache	6379
Celery Worker	Background task processing	—
VectorAI	Vector database (Action community)	8888

Start command: `docker-compose up --build`

10. Performance Metrics & Model Validation

Key Metrics

F1 Score

0.873

Test Set

Precision

0.977

Test Set

Recall

0.897

Test Set

CV F1

0.8808

Mean +/- 0.0017

Ensemble Model Weights

Model	Weight
XGBoost	0.25
LightGBM	0.25
Random Forest	0.15
Gradient Boosting	0.15
CatBoost	0.10
Extra Trees	0.10
Rule-Based Scoring	0.25 (of hybrid formula)

Feature Summary

- **35 Base Features:** amount, category, merchant, hour, day, cyclical time encoding (sin/cos), log transforms, Z-scores, rolling statistics (mean, std, count), interaction features (category×amount, hour×day)
- **3 Unsupervised Scores:** Isolation Forest score, LOF score, OCSVM score
- **1 Rule Score:** heuristic-based anomaly indicators
- **Feature scaling:** StandardScaler normalization

11. Complete Workflow Summary

End-to-End User Journey

1. **Authentication** — User signs up or logs in. JWT stored in localStorage. Protected routes guard all pages.
2. **CSV Upload** — User drag-and-drops a CSV file onto the Landing page's UploadZone. File sent to backend via POST /api/upload. Celery worker processes asynchronously with progress polling.
3. **ML Detection** — 48 features engineered per transaction. 10-model ensemble runs anomaly detection. Hybrid scoring combines ML predictions with rule-based heuristics.
4. **Map Display** — Results shown on interactive Marauder's Map SVG. Footprints animate, anomalies pulse with severity colors, category filter tabs available.
5. **Investigation** — User clicks anomaly for detail view. Three score gauges, Gemini-generated AI narrative with typewriter effect, ElevenLabs TTS audio narration.
6. **Actions** — Mark as 'Valid' or 'Mischief'. Export HTML report. Share via clipboard link. Use voice chat for natural language queries.

Data Flow Summary

User → React Frontend → FastAPI Backend → Redis/Celery → ML Ensemble → VectorAI/SQLite → Gemini AI → ElevenLabs TTS → User

Key Differentiators

10-Model Ensemble

Combines 6 supervised + 3 unsupervised + rule engine for robust detection

48 Features

Deep feature engineering including rolling stats, cyclical encoding, and interaction features

AI Narratives

Gemini generates contextual, engaging explanations for every anomaly

Interactive Map

SVG-based Marauder's Map with animated footprints and real-time updates

Voice Interface

Voice chat + TTS narration for accessibility and natural interaction

Dual Frontend

Two complete frontends (HP theme + OmniLedger tech theme) sharing one backend

"Mischief Managed."

The Marauder's Ledger v1.0 — AI-Powered Financial Anomaly Detection